Hop Data Vault 2.0 Plugin

Model-driven Data Vault and Business Vault on Apache Hop 2.18.1

The problem

Enterprise data warehouses built with hand-written pipelines drift from the model.

- Hashing rules change in one pipeline but not another
- Satellite CDC logic is copy-pasted across feeds
- Business Vault / SCD2 timelines are rebuilt in SQL or dbt without a shared contract
- Hybrid teams (Hop + dbt + SQL) cannot share one metadata picture

Result: slower delivery, inconsistent loads, and fragile consumption layers.

The approach

Model once. Generate loads and consumption layers.

1. Define sources, hubs, links, and satellites in a visual model (`.hdv`)
2. Validate structure and types before anything runs
3. Generate DDL and insert-only load pipelines automatically
4. Build Business Vault SCD2 tables from satellite history (`.hbv`)
5. Run controlled batch updates from workflow actions

The model is the contract — not a diagram that diverges from production.

Raw Data Vault layer

Hubs — core business entities (Customer, Product, Order)

Links — relationships between hubs

Satellites — descriptive history attached to hubs or links

Insert-only loading — new rows only when keys or attributes actually change; every batch shares one load timestamp.

Supports multi-active satellites, link satellites, load end dating, multi-source hubs, and sentinel rows (unknown / invalid).

Business Vault layer

Raw vault stores **technical** history (when the warehouse loaded a change).

Consumption often needs **functional** history (when the source says a value was true).

The plugin generates **SCD Type 2** Business Vault tables:

- `valid_from` / `valid_to` (or your column names)
- Open-ended sentinels for current rows
- Single-satellite or **multi-satellite merge** (e.g. Customer 360 from four satellites)

Saved as `.hbv` files, linked to the underlying `.hdv` model.

Hybrid warehouses

Not every raw vault table must be Hop-loaded.

Integration modes per hub, link, or satellite:

Mode	Meaning
Hop managed	Hop generates DDL and update pipelines (default)
External read-only	Table exists elsewhere (dbt, SQL, another ETL); Hop documents it for BV
Custom pipelines	Hop orchestrates your own `.hpl` files

Business Vault generation reads **metadata** (table layout, hash keys, attributes) — not Hop-generated DV pipelines. External tables participate fully in BV and catalog publish.

Model-driven operations

Operation	Purpose
Check model	Catch broken references and type mismatches early
Generate DDL	Preview CREATE/ALTER for the vault database
Debug	Open generated pipelines for inspection
Data Vault Update	Full raw vault batch load from a workflow
Business Vault Update	Rebuild BV tables from satellite history

Same validation engine in the GUI and in workflow actions.

Data catalog and lineage

- **Sources** live in the Hop data catalog (record definitions, field layouts, groups)
- Optional **publish** of target table layouts after DV or BV updates
- Foundation for downstream dimensional modeling and lineage (Marquez / OpenLineage planned)

Catalog + model = shared vocabulary between ingestion, vault, and consumption teams.

AI advisory (optional accelerator)

AI Help on the DV model toolbar:

- Analyze catalog sources and suggest hubs, links, satellites
- Propose layout and type mappings with review-before-apply
- Troubleshoot validation and integration errors

Requires a configured LLM in Hop GUI → Configuration → AI Assistant. Human modelers remain in control.

Quality and repeatability

The sample `project/` includes:

- Golden-dataset unit tests per suite
- Docker-based test runner (PostgreSQL, MySQL, SingleStore)
- Metrics JSON per Data Vault / Business Vault update run
- Orchestrated workflow covering basic vault, multi-active, link satellite, load end date, status tracking, multi-source hub, and **multi-satellite Business Vault (Customer 360)**

CI-friendly: fail fast on model checks or DDL drift.

Architecture (logical)

```
Sources (catalog / CRM)
        │
        ▼
  Raw Data Vault (.hdv)
  hubs · links · satellites
        │
        ├── Hop-managed loads
        ├── External read-only tables
        └── Custom .hpl orchestration
        │
        ▼
  Business Vault (.hbv)
  SCD2 · PIT · future bridges
        │
        ▼
  Dimensional / marts (roadmap)
```

One metadata chain from staging to consumption.

Who benefits

Role	Value
Architects	Governed hashing, naming, and load patterns; hybrid warehouse support
Modelers	Visual design, generated pipelines, BV without hand-written SQL
Operations	Workflow-driven batches, parallel execution, metrics, validation gates
Analytics / BI	Functional SCD2 timelines and standardized BV column names

Sample project highlights

Under `project/tests/`:

- **vault1** — classic hub / link / satellite
- **customer-phone** — multi-active satellite
- **customer-360** — four satellites merged into one BV SCD2 table
- **customer-360-external** — same BV model with **external read-only** DV satellites

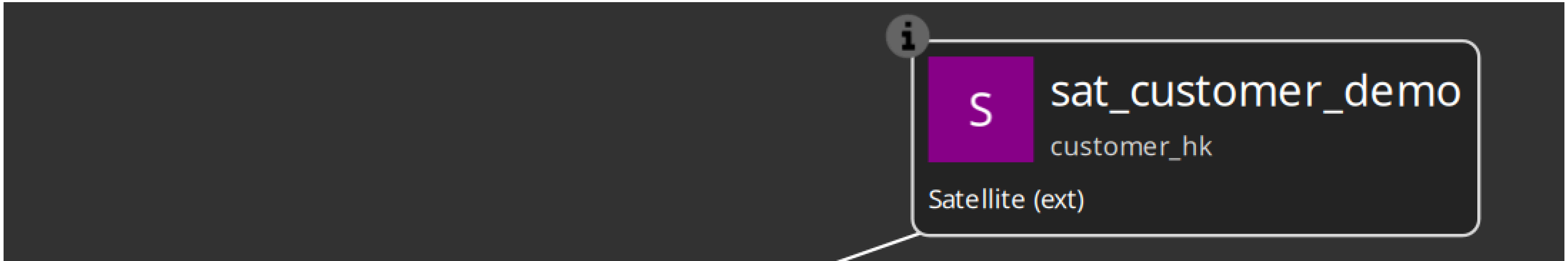
Integration modes (when to use which)

Hop managed — greenfield vault tables owned by Hop; full DDL + pipeline generation.

External read-only — dbt or another tool loads the raw vault; Hop models structure for BV, checks, and catalog. Canvas shows `(ext)`.

Custom pipelines — you maintain `.hpl` loaders; Hop runs them in the same orchestrator as generated pipelines. Canvas shows `(custom)`.

Choose per table — mixed models are supported.



Maturity and roadmap

Available today

- Raw DV modeling and loading
- Business Vault SCD2 (single and multi-satellite)
- PIT table type in BV model
- External / custom integration modes
- AI advisory

Planned

- Dimensional modeler (Kimball-style from DV/BV metadata)
- Business Vault field dictionary (naming / typing rules)
- Marquez / OpenLineage lineage export

Getting started

Audience	Start here
Modelers	docs/getting-started-modeler.adoc
Implementers	README.md and project/PROJECT.md
Reference	docs/README.md — full doc index

Open the `project/` folder as Hop project **`hop-data-vault`**, configure **CRM** and **Vault** connections, install plugin **0.0.14-SNAPSHOT**, and run `tests/run-tests.hwf`.

Summary

The Hop Data Vault plugin turns Data Vault 2.0 from a pile of pipelines into a **governed, model-driven platform**:

- Raw vault loads generated from `.hdv`
- Business vault SCD2 from `.hbv`
- External tables welcomed, not blocked
- Validation before execution
- Sample project proves it end-to-end

Model the warehouse. Let Hop generate the mechanics.